Department of Education

STEM Education
National Schools of Excellence
Teacher's Guide
Grade 12
Term 1

DOMAIN	Phylum
Science	Physics
Technology	
Engineering	
Mathematics	

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Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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SECRETARY'S MESSAGE

This Physics Syllabus is to be used exclusively by teachers of Physics to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Physics Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Physics before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Physics as a fundamental field of science.

This STEM Physics Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Physics.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Physics is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Physics to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Physics to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD

Secretary for Education

INTRODUCTION

Physics is the study of matter and its composition, how they interact to form new compounds through chemical reactions and the energy they generate as a result of these interactions. The concepts taught in Physics help us to understand these interactions at the molecular and atomic levels. Every substance, whether naturally occurring or artificially produced, consists of one or more elements of the periodic table.

Since Physics is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Physics.

The Grade 12 Physics Syllabus entails ten (6) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Physics is a highly specialized subject students undertaking Physics are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Teacher Guide are sequenced to guide teachers of Physics through the process of imparting basic chemistry knowledge and skills to students.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1 Biophysics	1. Introduction to Biophysics 2. Genomics – DNA Sequencing 3. Proteomics
	12.2 Bioelectronics	1. Applications of Bioelectronics
HEALTH AND MEDICINE	12.1 Nuclear Medicine	1. Introduction to Nuclear Medicine 2. Nuclear Medicine - Diagnosis 3. Nuclear Medicine - Therapy
ENERGY	12.1 Nuclear Energy	1. Introduction to Nuclear Energy 2. Nuclear Reactions – Fission and Fusion 3. Nuclear Fission Reactors
BATTERY TECHNOLOGY	12.1 Inductive Charging	1. Electromagnetism 2. Induction Charging 3. Hybrid Electric Vehicles

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Physics Bridging Syllabus students were introduced to the basic concepts and knowledge of Physics in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Physics as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

STEM DOMAIN

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM PHYSICS CONTENT OVERVIEW – GRADE 12, TERM 1

Table 1.1 Sequencing & Benchmarking STEM Physics Content Standards

Setting STEM Physics Content Standards for Progressive Learning & Assessment			
Thema	Biotechnology	Broad Content Standard	
Module 12.1: Biophysics		**12.1** Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1: Introduction to Biophysics	**12.1.1a:** Clearly define what biophysics and explain how it is as an integrated field of science. **12.1.1b:** Understand and differentiate the different applications in biophysics. **12.1.1c:** Differentiate between Genomics and Proteomics. **12.1.1d:** Describe the significant roles of Genomics and Proteomics in biophysics.	Students master the ability and knowledge to: • Recognize and understand biophysics as an integrated field of science. • Fundamental concept of Genomics (DNA Sequencing) and Proteomics (Protein Sequencing). • Sanger Sequencing. • Edman Degradation Process.	Research Project + Written Test
2: Genomics – DNA Sequencing	**12.1.2a:** Clearly define and describe Genomics and relate its concepts to physics. **12.1.2b:** Evidently distinguish between what a gene and genome is. **12.1.2c:** Highlight and articulate what DNA sequencing and Sanger Sequencing is. **12.1.2d:** List and describe the different ingredients used in Sanger Sequencing. **12.1.2e:** Explain and describe clearly the basic methodology and processes involved in performing Sanger Sequencing. **12.1.2f:** Understand and explain what Bioinformatics is.		

3: Proteomics	**12.1.3a:** Clearly differentiate between a protein and proteome. **12.1.3b:** Clearly outline and explain the relationship between Proteomics and Genomics. **12.1.3c:** Describe the process involved in Edman Degradation. **12.1.3d:** Compare the process of Edman Degradation and Sanger Sequencing.		
Module 12.1: Bioelectronics		**12.1** Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1: Applications of Bioelectronics	**12.2.1a:** define bioelectronics and its significance to biotechnology. **12.2.1b:** Clearly explain how the heart works and describe how the pacemaker is applied for patients with cardiac problems. **12.2.1c:** Clearly explain how the ear works and describe how the hearing aid assists patients with hearing problems.	Students master the ability and knowledge to: • Recognize and understand biophysics as an integrated field of science. • Fundamental concept of Genomics (DNA Sequencing) and Proteomics (Protein Sequencing). • Sanger Sequencing. • Edman Degradation Process.	Research Project + Written Test

Thema	Health and Medicine	**Broad Content Standard**	
Module 12.1	Nuclear Medicine	**12.1** Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.	
Topics	**Performance Indicators**	**Module Benchmark**	**Assessment Tasks**
1: Introduction to Nuclear Medicine	Students are expected to: **12.1.1a:** Define nuclear medicine as a combination of radiology, medicine and molecular biology.	Students' ability to master the knowledge and skills of; • Explaining the basic production and	Research Project + Written Test

	12.1.1b: Explain the general knowledge of physics utilized in nuclear medicine. **12.1.1c:** Explain what radiopharmaceuticals are, their production, and how it is administered to patients. **12.1.1d:** Assess and highlight the risks for involved in using nuclear medicine.	administering of radiopharmaceuticals. • Distinguishing the technologies and processes involved in nuclear diagnosis and therapy.	
2: Nuclear Medicine - Diagnosis	**12.1.2a:** Clearly define nuclear medicine diagnosis and emission tomography. **12.1.2b:** Highlight and generally explain the concept of radiation used in PET and SPECT scans. **12.1.2c:** Clearly differentiate the process and methods used in PET and SPECT scans. **12.1.2d:** Compare and contrast PET and SPECT scans and their effectiveness in medical diagnosis.		
3: Nuclear Medicine - Therapy	**12.1.3a:** Clearly define nuclear medicine therapy and explain in contrast to nuclear diagnosis. **12.1.3b:** Describe the physical and biochemical characteristics for choosing and administering a radionuclide. **12.1.3c:** Generally describe how radioactive Iodine therapy, treatment of bone metastases and radio immunotherapy is conducted.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	12.1	12.1	12.1	12.1	12.1
Biotechnology/ Module 12.1: Biophysics	**12.1** Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.	✓				
Biotechnology/ Module 12.2: Bioelectronics	**12.1** Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.		✓			
Health and Medicine/ Module 12.1 Nuclear Medicine	**12.1** Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.			✓		

THEMA: BIOTECHNOLOGY

Introduction

Biotechnology is a branch of biology and chemistry that employs mathematics and physics to develop tools and engineering insights for modern biology and biomedical research. It entails the study and application of living cells and cellular materials in the development of pharmacological, diagnostic, agricultural, environmental, and other products that promote human welfare and society. It's also utilized to examine and manipulate genetic information in animals in order to imitate and study human disease. Biotechnology is a science that is both fundamental and applied. Agriculture, health care and medical sciences, biomedical engineering, cell biology, immunology, seed technology, virology, plant physiology, forensics, industrial processing, and environmental management all benefit from the technologies it develops.

There are a number of career paths linked to the field of Biotechnology, such as: working in research labs, graduate training in biology and chemistry, and professional careers in the health sciences. If you are someone who enjoys STEM (particularly biology and chemistry), below are some best biotechnology careers that may be of interest to you (research on them for more information):

- Biomedical Engineer
- Biochemistry
- Medical Scientist
- Clinical Technician
- Microbiologist
- Process Development Scientist
- Biomanufacturing Specialist
- Business Development Manager
- Product Strategist
- Biopharma Sales Representative
- Medical Scientist
- Biotechnological Technician
- Epidemiologist
- Microbiologist
- Medical and Clinical Lab Technologist
- Biomanufacturing Specialist
- Bioproduction Specialist

MODULE 12.1: BIOPHYSICS

4-5 Weeks

Biophysics is a vibrant scientific field of study that comprises a variety of disciplines including mathematics, chemistry, engineering, material science and other life sciences. The primary aim of this field is to utilize and apply the principles and laws of physics to understand how biological systems work.

In this module, the basic concepts of biophysics will be covered that describes how this field acts as a bridge between the science domains (i.e., Physics, Biology and Chemistry). This will establish a clear understanding to then discuss its influence as a budding field of science, and its significant contributions to proteomics (protein science) and genomics (genetics) which have broader applications in biotechnology.

Prerequisites:

- Grade 11 Term 2, STEM Engineering (Thema Biotechnology)

Bridging Phyla(s):

- Grade 12 Term 1, STEM Biology (Thema Biotechnology)
- Grade 12 Term 1, STEM Chemistry (Thema Biotechnology)

Broad Content Standard

12.1 Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.

Content Benchmark

Students' ability to master the knowledge and skills of:

- Recognize and understand biophysics as an integrated field of science.
- Fundamental concept of Genomics (DNA Sequencing) and Proteomics (Protein Sequencing).
- Sanger Sequencing.
- Edman Degradation Process.

Topic 1: Introduction to Biophysics

Performance Indicators

Students are expected to;

12.1.1a: Clearly define what biophysics and explain how it is as an integrated field of science.
12.1.1b: Understand and differentiate the different applications in biophysics.
12.1.1c: Differentiate between Genomics and Proteomics.
12.1.1d: Describe the significant roles of Genomics and Proteomics in biophysics.

Review Exercise

Direct students to refer to their SRB and attempt the questions for review.
Resources: NIL

(Refer to TRB)

Class Discussion

In this activity, place the students into two groups to discuss about topic of DNA and Protein Sequencing. Allow the students to formulate discussions as they try to discover the benefits and drawbacks in using DNA and Protein Sequencing.

The following questions below can serve to guide the discussion.

1. What are the benefits of Gene and Protein Sequencing?
2. What are the drawbacks of Gene and Protein Sequencing?
3. Why are we not able to clone another human being?

(Refer to TRB)

Topic 2: Genomics – DNA Sequencing

Performance Indicators

Students are expected to;

12.1.2a: Clearly define and describe Genomics and relate its concepts to physics.

12.1.2b: Evidently distinguish between what a gene and genome is.

12.1.2c: Highlight and articulate what DNA sequencing and Sanger Sequencing is.

12.1.2d: List and describe the different ingredients used in Sanger Sequencing.

12.1.2e: Explain and describe clearly the basic methodology and processes involved in performing Sanger Sequencing.

12.1.2f: Understand and explain what Bioinformatics is.

Practical Experiment

Title: DNA Extraction and Sequencing

This activity is designed to be an inter-phyla project with Biology. It is a laboratory experiment where students are tasked to perform DNA extraction and sequencing on three different specimens but of the same family:

- Viola odorata
- Viola ignobilis
- Viola occulta

You can refer students to the Gr 12. Biology SRB - THEMA: BIOTECHNOLOGY, MODULE: Bioinformatics for clear instructions.

(Refer to the Gr 12. Biology SRB)

Topic 3: Proteomics

Performance Indicators

Students are expected to;

12.1.3a: Clearly differentiate between a protein and proteome.

12.1.3b: Clearly outline and explain the relationship between Proteomics and Genomics.

12.1.3c: Describe the process involved in Edman Degradation.

12.1.3d: Compare the process of Edman Degradation and Sanger Sequencing.

Review Exercise

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Module Quiz/Test

Prepare review questions and key concepts from each topic and hand to students to attempt.

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Biophysics	**12.1** Conceptualize the significant contribution of physics in biophysics and identify its practical concepts that can be applied to solve problems in the real world.	**1.** Introduction to Biophysics **2.** Genomics - DNA Sequencing **3.** Proteomics	Project + Written Quiz/Test

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
3. Proteomics	Written Quiz	What is Biophysics?	2.5
		How is proteomics and genomics related?	2.5
		What is the difference between a gene and a genome?	4
		List at least three applications of biophysics and briefly explain them.	6
		List the main ingredients used in Sanger sequencing.	2.5
		Briefly describe the Edman Degradation process.	2.5
		Subtotal	**20**
Note that this is similar to the Experiment given in GR 12 STEM Chemistry; THEMA: BIOTECHNOLOGY			
Module 12.1 (All three Topics)	Project/Experiment	Purify Protein Sequencing	20
		Determine the amino acid sequence of a protein	20
		Check if the protein sequence exist on a database of protein sequences online.	10
		Subtotal	**50**
Module 12.1 (All three Topics)	Written Test *(Multiple Choices, Short Answers, Calculations, etc)*	Correctly answers all long questions	10
		Correctly answers all multiple choice	15
		Correctly answers short answer questions	5
		Subtotal	***30***
		Module TOTAL	**100**

MODULE 12.2: BIOELECTRONICS

4-5 Weeks

Bioelectronics utilizes electronic devices to monitor basic biological functions and is particularly used to assist biological systems in humans that are diagnosed to be defective or underperforming. These electronic devices help to modulate, initiate and trigger the response that will otherwise naturally be triggered by biological processes.

In this module, we will discuss some key applications of bioelectronics that are commonly used that will illuminate the mindset of students to investigate into how the concepts of physics can have biological applications.

Prerequisites:

- NIL

Bridging Phyla(s):

- NIL

Broad Content Standard

12.1 Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.

Content Benchmark

Students' ability to master the knowledge and skills of:

- Recognizing and identifying bioelectronics that are currently used.
- Explaining the physics related to using bioelectronics in current real-life applications.

Topic 1: Applications of Bioelectronics

Performance Indicators
Students are expected to;
12.2.1a: define bioelectronics and its significance to biotechnology.
12.2.1b: Clearly explain how the heart works and describe how the pacemaker is applied for patients with cardiac problems.
12.2.1c: Clearly explain how the ear works and describe how the hearing aid assists patients with hearing problems.

Assignment
Prepare the Assignment and hand to students to attempt
Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.2: Bioelectronics	**12.1** Determine and highlight the role of bioelectronics that assist in creating devices for improving human deficiencies.	**1.** Applications of Bioelectronics	Assignment + Written Quiz

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1. Applications of Bioelectronics	Written Quiz	What is Bioelectronics?	2
		Explain how the heart is a natural pacemaker	2
		Describe why and how the cardiac pacemaker is used	5
		What is the difference between an ICD and CRT?	3
		Describe how sound travels through different parts of the human ear from the outer ear to the brain.	3
		Describe how the hearing aid works	5
		Subtotal	**20**
1. Applications of Bioelectronics	Assignment *(Research)*	Two bioelectronic devices identified	2
		Establishes the problem and how the device resolves it	3
		Clear explanation of how each device works	10
		Include sources and references	5
		Relevant use of diagrams to illustrate how th bioelectronic device works	5
		Subtotal	***25***
		Module TOTAL	**55**

THEMA: HEALTH AND MEDICINE

Introduction

The study of the human mind and body, how they function, and how they interact - not only with one another but also with their surroundings - has always been critical to human well-being. With the advancement of modern medicine, research on new therapies and preventative medicine has exploded, and a network of disciplines, including genetics, psychology, and nutrition, is working to improve human health.

MODULE 12.1: NUCLEAR MEDICINE

4-5 Weeks

Nuclear medicine is the science and clinical practice that involves the usage of radioactive compounds called radiopharmaceuticals (or more commonly called radiotracers) that are administered into the body. The radiation emitted is captured by certain medical instruments and assist nuclear medical practitioners to identify different types of diseases that can range from cancer, heart diseases and neurological disorders.

In this module, we will discuss the Physics associated with Nuclear Medicine. We will begin by understanding the nucleus of atoms at the sub-atomic level and how it relates to certain elements and chemical compounds becoming radioactive. This will serve as the basis to understanding radiotracers used in nuclear medicine and will assist students to understand how these instruments administer and use radiotracers to help identify and prevent the spread of the medical conditions mentioned above.

The main focus of Topic 1 is on Radionuclides used in Nuclear Medicine. Topic 2 focusses on Imaging modalities (SPECT and PET Scans). Topic 3 focusses on Therapeutic applications emphasizing on serious treatments such as cancer treatment and bone metastasis.

Prerequisites:

- Grade 11 Term 3, STEM Physics (Thema: Health and Medicine)

Bridging Phyla(s):

- NIL

Broad Content Standard

12.1 Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.

Content Benchmark

Students' ability to master the knowledge and skills of;

- Explaining the basic production and administering of radiopharmaceuticals.
- Distinguishing the technologies and processes involved in nuclear diagnosis and therapy.

Topic 1: Introduction to Nuclear Medicine

Performance Indicators

Students are expected to;

12.1.1a: Define nuclear medicine as a combination of radiology, medicine and molecular biology.

12.1.1b: Explain the general knowledge of physics utilized in nuclear medicine.

12.1.1c: Explain what radiopharmaceuticals are, their production, and how it is administered to patients.

12.1.1d: Assess and highlight the risks for involved in using nuclear medicine.

Review Exercise

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Topic 2: Nuclear Medicine - Diagnosis

Performance Indicators

Students are expected to;

12.1.2a: Clearly define nuclear medicine diagnosis and emission tomography.

12.1.2b: Highlight and generally explain the concept of radiation used in PET and SPECT scans.

12.1.2c: Clearly differentiate the process and methods used in PET and SPECT scans.

12.1.2d: Compare and contrast PET and SPECT scans and their effectiveness in medical diagnosis.

Review Exercise

Direct students to refer to their SRB and attempt the questions for review.

Resources: NIL

(Refer to TRB)

Topic 3: Nuclear Medicine – Therapy

Performance Indicators

Students are expected to;

12.1.3a: Clearly define nuclear medicine therapy and explain in contrast to nuclear diagnosis.

12.1.3b: Describe the physical and biochemical characteristics for choosing and administering a radionuclide.

12.1.3c: Generally describe how radioactive Iodine therapy, treatment of bone metastases and radio immunotherapy is conducted.

Class Discussion / Debate

Allow students to group and prepare them to discussion/debate on the use of Nuclear Medicine.

Resources: NIL

(Refer to TRB)

SAMPLE ASSESSMENT

Sample Assessment Methods

- Group Assignment
- Presentation
- Projects
- Written or Practical Quiz
- Written or Practical Test
- Practical Assignment
- Case Studies

Sample Assessment Tasks

Module	Content Standard	Topic	Assessment Tasks
12.1: Nuclear Medicine	12.1 Learn, recognize and comprehend the physics involved in using radiopharmaceuticals and the different nuclear emission tomographs used for medical diagnosis and therapy.	**1.** Introduction to Nuclear Medicine **2.** Nuclear Medicine – Diagnosis **3.** Nuclear Medicine – Therapy	Written Quiz + Written Test

Sample Assessment Marking Guide

Topic	Assessment Task	Assessment Criteria	Total Marks
1. Introduction to Nuclear Medicine	Written Quiz	What is meant by “nuclear” in nuclear medicine?	2
		What is medical Imaging and its relation to nuclear medicine?	2
		What are radiopharmaceuticals? List at least three different examples	4
		List at least one advantage and disadvantage of using nuclear medicine.	2
		Subtotal	**10**
Module 12.1 (All three Topics)	Written Test *(Multiple Choices, Short Answers, Calculations, etc)*	Correctly answers all long questions	20
		Correctly answers all multiple choice	20
		Correctly answers short answer questions	10
		Subtotal	50
		Module TOTAL	**60**

PLANNING, REPORTING & RECORDING (PERFORMANCE CRITERIA)

Assessment Planning

Assessment Domains	Modules 6	Total Marks	% Weight of Modules
Theory Knowledge & skills – **Test**	Test (50)	/50	**20**
Application of knowledge – **Assignments and projects**	Assign.1 (30)	/30	**30**
End of Term **Exam**	(25 marks)	/25	**50**
		Total marks for term one	**100 marks**

Assessment Recording

Percentage Range	Achievement Level
85 -100	Very High Achievement (VHA)
65-84	High Achievement (HA)
50-64	Satisfactory Achievement (SA)
25-49	Low Achievement (LA)
0-24	Below Minimum Achievement (BMA)

#	Names	Assessment Task								Total Score (/300)	Percentage Weight (%)				Achievement Level [VHA, HA, SA,LA]
		Tests		**Assignments**				Project	**Exam (/50)**		**T (20%)**	**A + P (30%)**	**Exam (50%)**	**Ttl CA**	
		1	**2**	**1**	**2**	**3**	**4**	1							
1	Bill Joe	45	47	20	21	22	23	55	45	***278***	18.4	25.88	45	***89.28***	VHA
2	Cara Den	20	25	15	16	17	18	25	30	***166***	9	18.22	30	***57.22***	SA
3	Lea May	9	15	5	6	7	8	30	25	***105***	4.8	9.10	25	***38.9***	LA

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